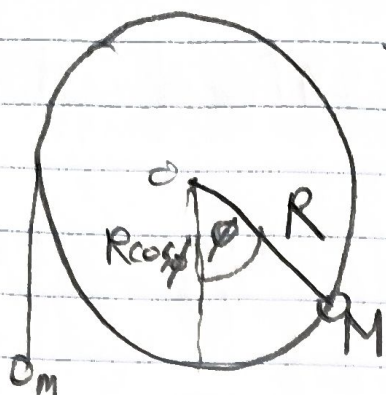


3d)



Take the zero PE to be at the center of the wheel

Displacing M by angle ϕ causes m to move downward by distance $R\phi$ since the string is inextensible

$$U_M = -MgR \cos \phi$$

$$U_m = -mgR\phi$$

is inextensible

$$U(\phi) = -MgR \cos \phi - mgR\phi = -gR(M \cos \phi + m\phi)$$

3b)

Critical points occur at ϕ where $\frac{dU}{d\phi} = 0$

$$U'(\phi) = -gR(-M \sin \phi + m) = 0$$

$$M \sin \phi = m$$

$$\sin \phi = \frac{m}{M} < 1$$

Critical points exist when

$$\frac{m}{M} < 1$$

other fixed point

$$\phi^* = \pi - \sin^{-1}\left(\frac{m}{M}\right)$$

$$\cos \phi^* = -\cos\left(\sin^{-1}\left(\frac{m}{M}\right)\right)$$

3c)

$$\sin \phi^* = \frac{m}{M}$$

$$\phi^* = \sin^{-1}\left(\frac{m}{M}\right)$$

$$\frac{d^2U}{d\phi^2} = -gR(-M \cos \phi) = MgR \cos \phi$$

$$\left. \frac{d^2U}{d\phi^2} \right|_{\phi^*} = -MgR \sqrt{1 - \frac{m^2}{M^2}}$$

(Unstable) < 0

$$\left. \frac{d^2U}{d\phi^2} \right|_{\phi^*} = MgR \sqrt{1 - \sin^2\left(\sin^{-1}\left(\frac{m}{M}\right)\right)} = MgR \sqrt{1 - \left(\frac{m}{M}\right)^2} > 0$$

Since $\frac{m}{M} < 1$, $\left. \frac{d^2U}{d\phi^2} \right|_{\phi^*} > 0$ thus ϕ^* is stable

The angle ϕ^* is where the mass M would oscillate around due to ϕ^* 's stability. This makes sense since the mass m's attachment to the rope causes tension, resisting mass M's fall resulting in oscillations at some stable point.

3e Expand $U(\phi)$ in a Taylor series centred at $\phi^* = \sin^{-1}\left(\frac{m}{M}\right)$ where $\frac{m}{M} \leq 1$

$$\begin{aligned} U(\phi) &= U(\phi^*) + U'(\phi^*)(\phi - \phi^*) + \frac{U''(\phi^*)}{2}(\phi - \phi^*)^2 + \dots \\ &= -gR\left(M \cos\left(\sin^{-1}\left(\frac{m}{M}\right)\right) + m \sin^{-1}\left(\frac{m}{M}\right)\right) + 0 \\ &\quad + \frac{MgR}{2} \sqrt{1 - \left(\frac{m}{M}\right)^2} (\phi - \phi^*)^2 + \dots \end{aligned}$$

$$U(\phi) = U(\phi^*) + \frac{MgR}{2} \sqrt{1 - \left(\frac{m}{M}\right)^2} (\phi - \phi^*)^2$$

The coefficient of $(\phi - \phi^*)^2$ can be thought of as a spring constant of $\frac{MgR}{2} \sqrt{1 - \left(\frac{m}{M}\right)^2}$ which causes oscillations. The spring constant is real if the term in the square root $\frac{m}{M} \leq 1$. Thus releasing from $\phi = 0$ with $\frac{m}{M} \leq 1$ causes oscillations.